



Brain Tumor Segmentation Using UNet-Context Encoding Network

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Abstract. Glioblastoma is an aggressive type of cancer that can develop in the brain or spinal cord. Magnetic Resonance Imaging (MRI) is key to diagnosing and tracking brain tumors in clinical settings. Brain tumor segmentation in MRI is required for disease diagnosis, surgical planning, and prognosis. As these tumors are heterogeneous in shape and appearance, their segmentation becomes a challenging task. The performance of automated medical image segmentation has considerably improved because of recent advances in deep learning. Introducing context encoding with deep CNN models has shown promise for semantic segmentation of brain tumors. In this work, we use a 3D UNet-Context Encoding (UNCE) deep learning network for improved brain tumor segmentation. Further, we introduce epistemic and aleatoric Uncertainty Quantification (UQ) using Monte Carlo Dropout (MCDO) and Test Time Augmentation (TTA) with the UNCE deep learning model to ascertain confidence in tumor segmentation performance. We build our model using the training MRI image sets of RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021. We evaluate the model performance using the validation and test images from the BraTS challenge dataset. Online evaluation of validation data shows dice score coefficients (DSC) of 0.7787, 0.8499, and 0.9159 for enhancing tumor (ET), tumor core (TC), and whole tumor (WT), respectively. The dice score coefficients of the test datasets are 0.6684 for ET, 0.7056 for TC, and 0.7551 for WT, respectively.

Keywords: Glioblastoma · Segmentation · Deep neural network · Uncertainty · Monte Carlo dropout · Test time augmentation

1 Introduction

Glioblastoma (GBM) is the most common and aggressive malignant primary tumor of the central nervous system (CNS) in adults, with extreme intrinsic heterogeneity in appearance, shape, and histology [1]. Patients diagnosed with the most aggressive type of brain tumor have a median survival time of two years or less [2]. Accurate brain tumor segmentation is important not only for treatment planning but also for follow-up evaluation [3]. Manual brain tumor segmentation is time-consuming, less efficient, and prone to error [4, 5]. Therefore, it is desirable to have a computer-aided

image-based robust automatic brain tumor segmentation system. Different strategies for brain tumor segmentation have been investigated, including threshold-based, region-based, and traditional machine learning-based methods [6–9]. However, those methods are limited because of the complex mathematic model or difficult hand-crafted feature extraction. Recent deep learning (DL) models-based automated tumor segmentation systems have demonstrated considerable performance improvements [3, 10]. DL has made it feasible to build large-scale trainable models that can learn the best features for a specific task [11]. To achieve successful tumor segmentation, DL models often require many training examples. It is a very challenging task to build an accurate deep learning model because of the lack of biomedical and bioimaging datasets. Therefore, proper regularization and hyper-parameter tuning are required for developing an efficient DL network.

Inspired by the popular deep learning architecture known as UNet [12] and the concept of context encoding designed for semantic segmentation [13], we implement a state-of-the-art deep UNet-Context Encoding (UNCE) framework for automatic brain tumor segmentation. Additionally, we compute the uncertainty using a combination of Monte Carlo dropout (MCDO) and test time augmentation (TTA) of data to improve the overall performance and to obtain a confidence measure in the brain tumor segmentation outputs [14].

2 Methods

2.1 Data Descriptions

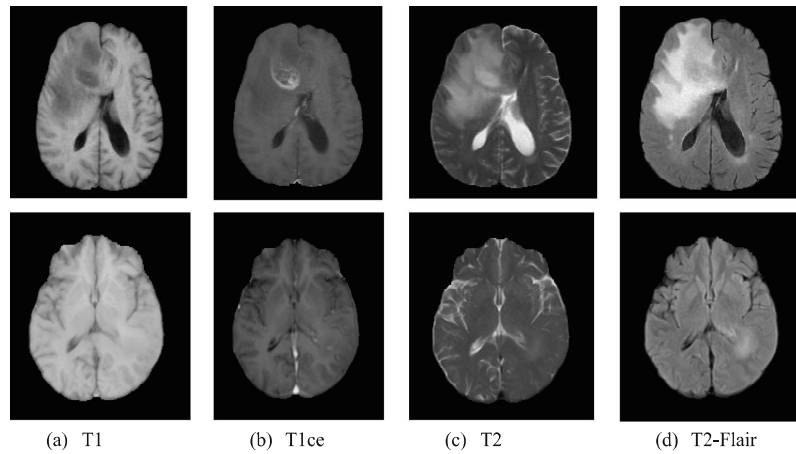


Fig. 1. Examples of four different MRI modalities of two different training samples: (a) T1, (b) T1ce, (c) T2 and (d) T2-FLAIR.

The RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021 dataset is obtained from multiple different institutions under standard clinical conditions

[1]. Different institutions used different equipment and imaging protocols which resulted in a vastly heterogeneous image quality reflecting diverse clinical practices across different institutions. Ground truth annotations of every tumor sub-region for brain tumor segmentation were approved by expert neuroradiologists [15–18]. The BraTS 2021 training dataset consists of 1251 cases with the ground truth labels. In the validation phase, there are 219 cases are provided without any associated ground truth. Each patient case has four MRI modalities: T1-weighted (T1), T1-weighted contrast enhancement (T1ce), T2-weighted (T2), and T2-weighted fluid-attenuated inversion recovery (T2-FLAIR). All modality sequences are co-registered, skull-stripped, denoised, and bias-corrected [19]. Image size is 240 X 240 X 155 for each imaging modality. Tumors have different sub-tissues: necrotic (NC), peritumoral edema (ED), and enhancing tumor (ET). Figure 1 shows the images of four MRI modalities (T1, T1ce, T2 and FLAIR) of two different training examples.

2.2 Network Architecture

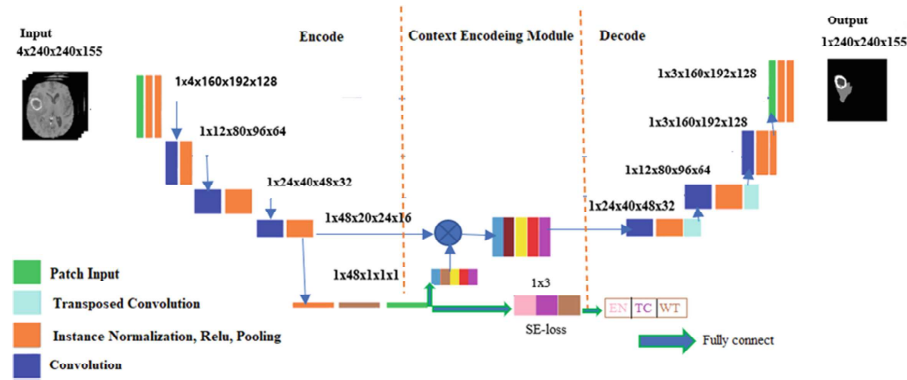


Fig. 2. Overview of the UNet-Context Encoding (UNCE) Network.

The UNet is a convolutional network architecture used for fast and precise segmentation of images. The bottleneck layer of UNet captures the global semantic context features of the scene with rich contextual information. In this work, we have implemented an UNCE network architecture that integrates multiple volumetric MRI processing tasks. Inspired by the work of context encoding network [13], the UNCE architecture is substantially augmented for brain tumor segmentation.

An overview of the UNCE deep learning method for tumor segmentation is shown in Fig. 2. The UNCE captures global texture features using a semantic loss to provide regularization in training. The architecture consists of encoding, context encoding, and decoding modules. The encoding module extracts high-dimensional features of the input. The context encoding module produces updated features and a semantic loss to regularize the model by ensuring all segmentation classes are represented. The decoding module reconstructs the feature maps to produce segmentation masks as output. Figure 2 presents

a detailed architecture of the proposed UNCE model including the parameter settings used at each layer.

2.3 Implementation

The MRI scans provided for the competition are co-registered, skull-stripped, denoised and bias corrected. Since the dataset was collected from different institutions and MRI scanners, the intensities show substantial variations across examples. Consequently, we perform normalization of all examples to have zero mean and unit standard deviation. The dimension of each training sample is $240 \times 240 \times 155$. The size of the images is reduced by cropping to a size of $192 \times 160 \times 128$ to manage computational memory and cost. To generate additional training images, we augment data by adding uniform white noise with limited amplitude.

A critical feature of the proposed UNCE is the context encoding module, which computes scaling factors related to the representation of all classes. These factors are learned simultaneously in the training phase via the semantic loss error regularization, defined by L_{se} . The scaling factors capture global information of all classes, essentially learning to mitigate the training bias that may arise due to imbalanced class representation in data. To calculate the Semantic Error loss (SE-loss), we construct another fully connected layer with a sigmoid activation function upon the encoding layer, so that predict object classification in the image [13]. Accordingly, the final loss function consists of 2 terms: Subsequent paragraphs, however, are indented.

$$L = L_{dice} + L_{se} \quad (1)$$

where L_{dice} is a Dice calculated by the difference between prediction and ground truth, and L_{se} is the sematic loss.

Dice loss is computed as:

$$L_{dice} = 1 - DSC \quad (2)$$

where DSC is dice similarity coefficient [20]. The DSC is defined as,

$$DSC = \frac{2TP}{FP + 2TP + FN} \quad (3)$$

where TP, FP and FN are the numbers of true positive, false positive and false negative, respectively.

We use Adam optimizer [20] with initial learning rate of $lr_0 = 0.0001$ in training phase, and the learning rate (lr_i) is gradually reduced by following:

$$lr_i = lr_0 * \left(1 - \frac{i}{N}\right)^{0.9} \quad (4)$$

where i is epoch counter, and N is a total number of epochs in training.

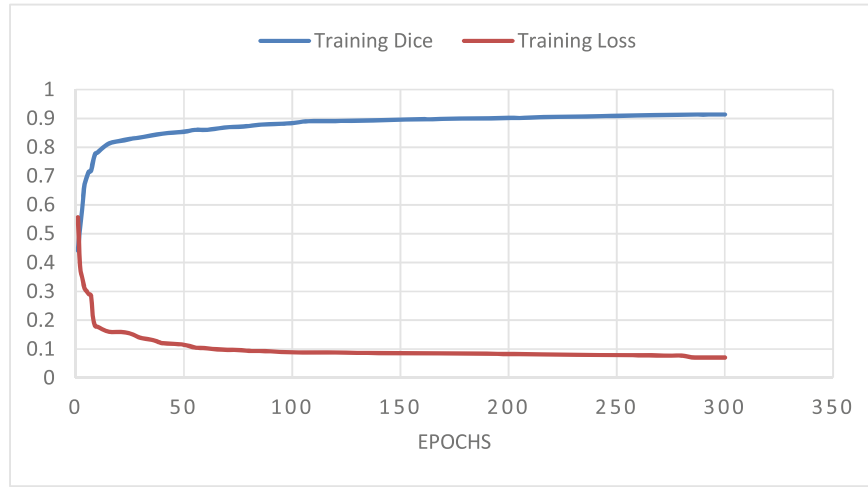


Fig. 3. Training loss and training dice vs number of epochs.

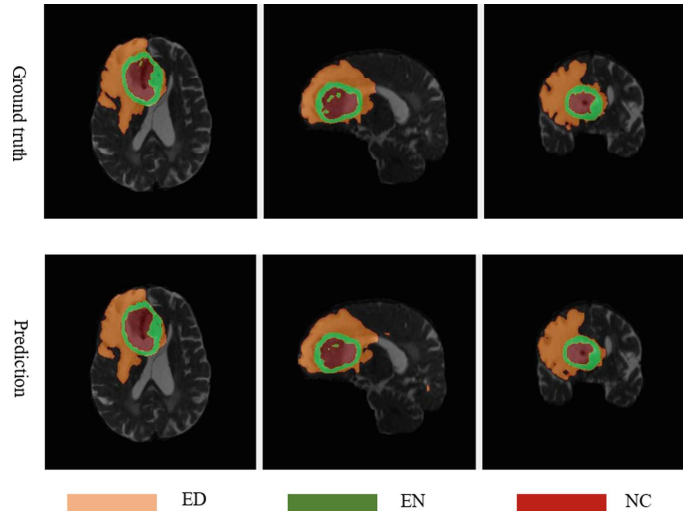


Fig. 4. T2 image overlaid with our prediction. From left to right: axial, coronal, sagittal view respectively.

2.4 Training

We implement the context-aware deep learning network in PyTorch and train the network on NVIDIA V100 HPC platform using the 2021 BraTS training dataset (1251 cases). To train the network, we use 80% of all training data and the remaining 20% of the training data is used to validate the trained model. We train the network using one of the regularization techniques known as a dropout. In this implementation, 30% of random weight was dropped to train the network. The same network was also trained without

applying the dropout. The UNCE architecture is trained for over 300 epochs and the best performing versions based on the validation set are retained for testing. Figure 3 shows the training loss and training soft dice curve of the UNCE for the BraTS 2021 dataset. Effective training of the network is observed by the monotonically decreasing loss and the corresponding increase in training dice score. The network achieves an overall dice score of 0.90 by the first 110 epochs and continues to further improve performance at a slower pace. At the end of 300 epochs, we get an average training loss of 0.0701 and training soft dice reaches 0.9131. Once the network is fully trained, the performance of the network is evaluated using the BraTS 2021 validation dataset (219 cases) utilizing the online submission process made available by the challenge organizers. Figure 4 presents an example segmented tumor and the corresponding ground truth of T2 images. The trained model segmented results are very close to the ground truth.

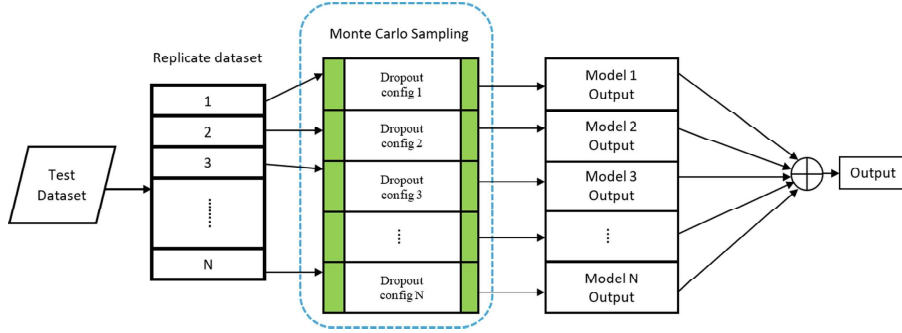


Fig. 5. Monte Carlo dropout computation framework

2.5 Uncertainty Measures

Bayesian probability theory offers mathematically grounded tools to reason about model uncertainty. However, Bayesian techniques with deep learning usually result in models with prohibitive computational costs. Therefore, it may be prudent to utilize methods that are able to approximate the predictive posterior distribution without changing either the models or the optimization [22]. For instance, Monte Carlo Dropout (MCDO) utilizes the dropout layers within a deep neural network at test time to conduct Monte Carlo sampling of the parametric posterior to obtain an approximation of the predictive posterior distribution. We adopt this technique for the UNCE as follows: 1) activate dropout layers in UNCE at test time. 2) Replicate each testing image y , N times and pass each through the dropout enabled UNCE. This essentially runs each replication through a different version of UNCE generated by random dropout masks. 3) obtain the N different predictions $P_n(y)$, where each prediction is a vector of SoftMax scores for the C classes. 4) Then we compute the average prediction score [23] for the N samples as follows.

$$P_N(y) = \frac{1}{N} \sum_{n=1}^N P_n(y) \quad (5)$$

In the MCDO technique, we turn on the dropout during the evaluation. Additionally, we conduct test time augmentation of inputs to further increase the sample size and to capture epistemic uncertainty in the predictions. The final segmentation results are obtained by averaging over all the outputs obtained after applying a combination of MCDO and TTA to a given testing example as shown in Eq. (5). Figure 5 shows the framework of the MCDO method. The output from each Monte Carlo sampling model is fused to get the final segmented results.

3 Results and Discussion

Table 1. Performance of the BraTS 2021 validation data for three models.

UNCE Model	Statistical parameter	Dice Score			Hausdorff95		
		ET	TC	WT	ET	TC	WT
No Dropout	Mean	0.7696	0.8281	0.9121	20.24	12.10	4.71
	std	0.2489	0.2395	0.0762	77.34	50.10	7.64
	Median	0.8532	0.9278	0.9348	2.24	2.24	2.83
	25quantile	0.7713	0.8324	0.8894	1.41	1.41	1.73
	75quantile	0.9016	0.9544	0.9547	3.16	5.15	4.64
Dropout	Mean	0.7720	0.8465	0.9141	20.51	7.03	4.07
	std	0.2555	0.2011	0.0746	77.36	26.43	5.65
	Median	0.8589	0.9291	0.9350	2.24	2.00	2.45
	25quantile	0.7870	0.8376	0.8947	1.41	1.41	1.41
	75quantile	0.9052	0.9570	0.9575	3.19	5.00	4.24
MCDO + TTA	Mean	0.7786	0.8499	0.9159	20.26	6.71	4.35
	std	0.2501	0.2031	0.0752	77.35	25.97	6.98
	Median	0.8606	0.9310	0.9366	2.00	2.00	2.45
	25quantile	0.7837	0.8321	0.8980	1.41	1.41	1.41
	75quantile	0.9084	0.9579	0.9588	3.00	5.01	4.24

We first obtain the best performing UNCE models using the training dataset provided by the RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021 organizer. We use three models (Model without dropout, Model applying dropout and model using combined MCDO and TTA) to generate the segmented results. These models are then evaluated using the validation dataset through the online evaluation tool Table 1. Shows the DSC, and Hausdorff distance of the validation dataset of three such models for 219 examples. The DSC quantifies the similarity between tumor segmentation and ground truth and Hausdorff distance measures how far two subsets (tumor segmentation and ground truth) of a metric space are from each other. The dice scores

of the model without dropout are 0.7698, 0.8281 and 0.9121 for enhancing tumor (ET), tumor core (TC) and whole tumor (WT) respectively. The hausdorff95 distances of the same model are 20.24 for ET, 12.10 for TC and 4.71 for WT. The model that is trained by applying dropout in each layer of the network show improvements in the dice scores and hausdorff95 distances for all three tumor cases. The third model uses MCDO and TTA to further improve performance through the uncertainty quantification strategy discussed in Sect. 2.5. For this experiment, we set the Monte Carlo sample number $N = 100$ and obtain the average of corresponding sample outputs to form the final segmentation results. The evaluation results show a further improvement of dice scores compared to the other two models.

The final model was tested by the BraTS challenge organizers. Following the requirement of the challenge, we prepared a Docker image of our model. In the test phase, 570 MRI images were segmented to evaluate the model performance. Note among the 570 test examples, 87 were not segmented successfully. Table 2 shows the test results for our model.

Table 2. Performance of the BraTS 2021 Test data.

UNCE Model	Statistical parameter	Dice Score			Hausdorff95		
		ET	TC	WT	ET	TC	WT
MCDO + TTA	Mean	0.6684	0.70562	0.7551	74.26	74.36	62.77
	std	0.3527	0.38792	0.3544	145.76	144.47	132.87
	Median	0.8424	0.9239	0.9262	2.23	2.23	3
	25quantile	0.6582	0.5461	0.8148	1.41	1	1.41
	75quantile	0.8984	0.9601	0.9588	6.38	14.08	11.08

4 Conclusion

In this work, we use a UNet-context encoding 3D deep learning based method for brain tumor segmentation. The model takes advantage of context encoding, which captures global context features. Semantic error loss is used to regularize the model that helps to manage the class specific sample bias that exists within the tumor tissue segmentation task. BraTS2021 training dataset is used to generate and train several versions of the proposed UNet-Context Encoding (UNCE) model and evaluate three representative versions using the validation dataset provided by the BraTS20201 organizers. The results show that the UNCE version combined with the Monte Carlo Dropout (MCDO) and Test Time Augmentation (TTA) based uncertainty quantification framework yields the best performance.

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